



PARAMETER ESTIMATION USING KALMAN FILTERS WITH CONSTRAINTS

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We suggest incorporating dynamical information such as locations of unstable fixed points into parameter estimation algorithms in order to improve the method of reconstructing dynamics from time series data. We show how the process of reconstruction using nonlinear filters such as the extended Kalman filter can be easily modified to take advantage of the additional information. We demonstrate the methods using data from two systems exhibiting chaotic dynamics — the Chua circuit and Chen's equations. In both cases we find the models reconstructed using constraints that better approximate the unstable fixed point structure of the underlying systems.

Keywords: Kalman filter; nonlinear systems; parameter fitting; unstable fixed points.

1. Introduction

This paper is concerned with a dynamical-systems approach to nonlinear systems identification [Abarbanel, 1995; Kantz & Schreiber, 1997]. The approach is based on Takens Embedding Theorem [Takens, 1981] which allows reconstruction of an equivalent phase space to the original (unknown) system from observations of the system in the form of a time series. The dynamics in this reconstructed phase space can be approximated using, if necessary, nonlinear functions. Typical architectures for time series prediction include but are not limited to local polynomials [Abarbanel, 1995], neural networks [Small & Tse, 2002] and radial basis functions [Judd & Mees, 1995, 1998]. The estimation of model parameters in these architectures can result in nonlinear optimization problems.

In recent times the extended Kalman filter (EKF) algorithm has been suggested as an alternative tool for solving these optimization problems to more traditional methods such as steepest descent. In [Haykin, 2001] the various chapters

show the benefits of Kalman filtering for parameter estimation when neural networks are the approximating functions. Puskorius and Feldkamp [1994] had earlier pointed out how the parameter-based EKF training algorithm can be naturally incorporated into nonlinear control architectures. Burgmeier [1995], Walker and Mees [1998] and Walker [2002] explained how all parameters in a radial basis function model can be estimated using the EKF algorithm.

The problem of reconstructing dynamics of systems by estimating parameters of nonlinear functions can be thought of as “black-box” modeling. One can imagine that more “grey-box” modeling would produce in some sense improved nonlinear models. Grey-box modeling can be thought of as an approach to reconstruction where additional information, other than the time series data, is used in the modeling process. For example, if the system is known to possess symmetry, one could select basis functions which preserve such symmetries. It might be expected that the resulting models would exhibit

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dynamical properties of the system more closely. Indeed in [Walker *et al.*, 2001] it was shown that basis functions which mimicked a fading memory assumption resulted in more accurate free-running models over standard radial basis function models. Instead of imposing additional structure to the basis functions an alternative “grey-box” approach to modeling is to include properties of the system as constraints in the parameter estimation optimization problems.

It is known that the (unstable) periodic orbits of chaotic systems play a significant role in the structure of attractors [Artuso *et al.*, 1990a, 1990b]. We suggest using the low-order periodic structure, in particular unstable fixed points (UFP), as additional constraints in the parameter estimation process. The resulting models in addition to providing a reasonable fit to the training data should also better capture the local dynamic behavior associated with the constraints.

The EKF is a natural environment in which to incorporate such constraints. The location of UFP’s and low-order periodic orbits can be determined from data [Auerbach *et al.*, 1987; Glover & Mees, 1993; So *et al.*, 1996; So *et al.*, 1997]. The purpose of this paper is to demonstrate the idea of using these UFP’s as constraints within the EKF framework. The errors in estimating the location of UFP’s can also be used to advise on levels of noise covariances within the EKF.

This paper is structured as follows: In Sec. 2 we discuss the nonlinear functions we use to reconstruct dynamics and illustrate the methods. We also briefly describe our favored method for locating UFP’s. In Sec. 3 we outline the extended Kalman filter algorithm in the context of parameter estimation and introduce the modifications required in order to use the Kalman filter with constraints. The new approach is demonstrated using data from two systems which can exhibit chaotic dynamics, namely, the well-known Chua circuit [Madan, 1993; Kennedy, 1993] and equations by Chen [Chen & Ueta, 1999]. The results are presented in Sec. 4 and the benefits of using UFP information as constraints in the EKF are shown. The paper closes in Sec. 5 with a short discussion and summary of the methods.

2. Nonlinear Models

A nonlinear dynamical system can be modeled in continuous time and discrete time. Continuous

time models are typically represented by differential equations whereas discrete time models can be described by maps. We shall concentrate on reconstructing nonlinear dynamics using discrete time maps. A model of a system with state $\mathbf{z} \in \mathbf{R}^L$ is described by

$$\mathbf{z}_{k+1} = \mathbf{F}[\mathbf{z}_k, \mathbf{a}] \quad (1)$$

where $\mathbf{a}$ represents the system parameters. The problem of modeling the dynamics from scalar time series requires the state $\mathbf{z}$ to be reconstructed. This is typically done using time delay embedding [Takens, 1981]. Given a scalar time series $\{y_t\}_{t=0}^N$ the state $\mathbf{z}_k$ can be represented by

$$\mathbf{z}_k = (y_{k-(m-1)\tau}, \dots, y_{k-\tau}, y_k) \quad (2)$$

where m is called the embedding dimension and τ is the time delay lag. There are a number of methods which can be used to determine an appropriate τ with a good prescription being to choose the first minimum of the average mutual information function [Fraser & Swinney, 1986; Abarbanel, 1995]. Similarly, a recognized method for selecting a suitable embedding dimension is false nearest neighbors [Kennel *et al.*, 1992; Abarbanel, 1995]. There are more general time delay embeddings referred to as nonuniform embedding where the lags are not multiples of a single time delay [Judd & Mees, 1998]. These nonuniform embeddings can give better representations of the system state but for the purposes of this paper we will use the standard embedding given in (2).

The dynamics $\mathbf{F}[\cdot]$ for a time delay embedding consist of a simple shift operator together with a (nonlinear) scalar-valued function f , i.e.

$$y_{k+1} = f(\mathbf{z}_k) \quad (3)$$

There are a number of choices to use as the function f including but not restricted to, local polynomial models, thin-plate splines, neural networks and radial basis functions. A class of models which have been particularly successful in capturing nonlinear dynamics from data is the so-called pseudo-linear models [Judd & Mees, 1995] which include a linear combination of linear and nonlinear basis functions. This is the class of models we will consider in this study, namely,

$$y_{k+1} = \sum_{i=1}^K \omega_i \phi(\|\mathbf{c}_i - \mathbf{z}_k\|) + \sum_{i=0}^m \alpha_i y_{k-i\tau} + \beta \quad (4)$$

The function $\phi(\cdot)$ is a radial basis function and ω_i , α_i and β are weight parameters to be determined. The $\mathbf{c}_i$ are the radial basis centers whose locations must also be determined. There are numerous choices for the form of the basis function with Gaussian's being widely used. In this paper we use the following basis function

$$\phi(u) = \frac{1}{1 + \cosh(-\delta u)}, \quad u \geq 0 \quad (5)$$

This function has similar behavior to Gaussians but can perhaps be more readily implemented using electronic components. The scale factor δ is another parameter requiring determination and could be a different value for each individual basis function. We will explain how the EKF can be used to estimate all unknown parameters of a pseudo-linear model but for ease of exposition will restrict the problem to determining the weight parameters in the examples.

2.1. Unstable fixed points

The steady-state or long term behavior of a dynamical system is of particular interest. The properties of unstable fixed points and unstable periodic orbits of a system play an important role in determining this behavior. An unstable fixed point of a discrete time dynamical system satisfies

$$\bar{\mathbf{z}} = \mathbf{F}[\bar{\mathbf{z}}, \mathbf{a}] \quad (6)$$

where $\bar{\mathbf{z}}$ denotes the UFP. The fixed points of a pseudo-linear model can be seen to satisfy

$$p = f(\mathbf{p}) \quad (7)$$

where $\mathbf{p} = (p, p, \dots, p) \in R^m$ is the fixed point location in reconstructed phase space. There have been a number of methods proposed in the literature for extracting dynamical information such as UFP locations. These include studying close returns [Auerbach *et al.*, 1987], or investigating properties of locally reconstructed models [Glover & Mees, 1993; So *et al.*, 1996; So *et al.*, 1997]. We use a modification of the Glover and Mees [1993] method to determine UFP's. In [Glover & Mees, 1993], a sliding window is applied across the time series and a linear model is reconstructed in each window. If the window is $\{y_1, y_2, \dots, y_{2m+1}\}$ then a linear model is

$$\sum_{i=k}^{k+m} c_i y_i = 1, \quad k = 1, \dots, 2m+1 \quad (8)$$

which can be written as

$$Y\mathbf{c} = \mathbf{1}_{m+1} \quad (9)$$

where $\mathbf{1}_{m+1}$ is a vector of ones. For each model we can calculate a fixed point as

$$p^{-1} = \sum_{i=1}^{m+1} c_i \quad (10)$$

For each k , these p_k are candidate fixed points of the system. In order to determine how meaningful each estimate is Glover and Mees form $r_k = \|\mathbf{v}_k - \mathbf{q}_k\|$ where $\mathbf{v}_k = (y_k, \dots, y_{k+m-1})$ and $\mathbf{q}_k = (p_k, \dots, p_k)$ and plot p_k versus $\log r_k$, or histograms of p_k . The fixed points should reveal themselves as definite spikes in these plots. This technique has proved quite effective in locating UFP's both on and off the attractor. The method written and presented above and in [Glover & Mees, 1993] uses neighbors in time but can readily be modified to use neighbors in reconstructed phase space. In the sequel we have used neighbors in space to determine candidate UFP's in our examples. The methods of [Glover & Mees, 1993] were also extended in [So *et al.*, 1996] and [So *et al.*, 1997] to extract higher order unstable period orbits from time series data.

3. Extended Kalman Filter and Parameter Estimation

The Kalman filter and its variant to deal with nonlinear systems, the extended Kalman filter, are statistical state estimators. The algorithms are typically used to achieve noise reduction in signal processing but can be applied to the problem of parameter estimation. The system model required by the Kalman filter for this particular application is

$$\begin{aligned} \mathbf{a}_t &= \mathbf{a}_{t-1} \\ y_t &= f(\mathbf{z}_t, \mathbf{a}_t) + n_t \end{aligned} \quad (11)$$

where $\mathbf{a}_t$ represents the model parameters. These can be the pseudo-linear weights, the location of the centers and the values of radial basis function scale parameters. There is no additive noise term in the evolution of $\mathbf{a}_t$ as the parameters are assumed to be stationary. The second equation is the pseudo-linear model used to predict the time series values y_t and $\mathbf{z}_t$ is the embedded time series data. The noise term n_t can be thought of as the fitting error and is assumed to be from a Gaussian distribution, i.e. $n_t \sim N(0, R)$. The parameters can appear in the function f nonlinearly and so

an EKF is required to estimate $\mathbf{a}$. The parameters are initialized with $\mathbf{a}_0 \sim N(\hat{\mathbf{a}}_0, \mathbf{P}_{\mathbf{a}_0})$ where $\hat{\mathbf{a}}_0 = E[\mathbf{a}_0]$ and $\mathbf{P}_{\mathbf{a}_0} = E[(\mathbf{a}_0 - \hat{\mathbf{a}}_0)^T(\mathbf{a}_0 - \hat{\mathbf{a}}_0)]$. The EKF update equations for the system (11) takes the form [Jacobs, 1993; Walker & Mees, 1998; Haykin, 2001]

time-update equations

$$\begin{aligned}\hat{\mathbf{a}}_t^- &= \mathbf{a}_{t-1} \\ \mathbf{P}_{\mathbf{a}_t}^- &= \mathbf{P}_{\mathbf{a}_{t-1}}\end{aligned}$$

measurement-update equations

$$\begin{aligned}\mathbf{K}_t &= \mathbf{P}_{\mathbf{a}_t}^- \mathbf{C}_t^T (\mathbf{C}_t \mathbf{P}_{\mathbf{a}_t}^- \mathbf{C}_t^T + \mathbf{R})^{-1} \\ \hat{\mathbf{a}}_t &= \hat{\mathbf{a}}_t^- + \mathbf{K}_t [\mathbf{y}_t - f(\hat{\mathbf{z}}_t, \hat{\mathbf{a}}_t)] \\ \mathbf{P}_{\mathbf{a}_t} &= (\mathbf{I} - \mathbf{K}_t \mathbf{C}_t) \mathbf{P}_{\mathbf{a}_t}^-\end{aligned}$$

where

$$\mathbf{C}_t = \left. \frac{\partial f(\mathbf{z}, \mathbf{a})}{\partial \mathbf{a}} \right|_{\hat{\mathbf{a}}_t}$$

The matrix $\mathbf{K}_t$ is referred to as the *Kalman gain*. The terms $[\mathbf{y}_t - f(\hat{\mathbf{z}}_t, \hat{\mathbf{a}}_t)]$ are known as the innovations.

3.1. Constrained Kalman filtering

The Kalman filter and its nonlinear variant have proved to be successful in estimating the parameters of nonlinear models which capture the dynamical behavior of systems exhibiting chaos [Walker & Mees, 1998; Patel & Haykin, 2001]. In [Walker *et al.*, 2001] it was shown that selecting basis functions which include basic properties of the system can lead to improved model reconstruction. In this paper rather than impose properties on the selected basis functions we suggest treating important properties of the system as constraints to be met as accurately as possible. The setup of the Kalman filter is readily amenable to including such constraints. The constraints, or properties, we will try to meet are the location of UFP's of the system. We can do this by modifying the observation model of (11) to include the model's approximation to the UFP's. The system model for applying the Kalman Filter with constraints, referred to in the sequel as the KFC algorithm becomes

$$\begin{aligned}\mathbf{a}_t &= \mathbf{a}_{t-1} \\ y_t &= f(\mathbf{z}_t, \mathbf{a}_t) + \mathbf{n}_t \\ 0 &= p - f(\mathbf{p}, \mathbf{a}_t) + m_t\end{aligned}$$

where the new term includes $m_t \sim N(0, R_p)$ where R_p reflects the accuracy of the knowledge of the fixed point location. The value R_p is set so as to

be guided by the histograms which result from the UFP locator analysis described in Sec. 2.1. The filter update equations are essentially unchanged but the matrices and vectors in the measurement update equations now account for $N_p + 1$ observation functions, where N_p are the number of UFP constraints we try to meet. Explicitly the measurement update equations become

measurement-update equations

$$\begin{aligned}\mathbf{K}_t &= \mathbf{P}_{\mathbf{a}_t}^- \mathbf{C}_t^T (\mathbf{C}_t \mathbf{P}_{\mathbf{a}_t}^- \mathbf{C}_t^T + \mathbf{R}')^{-1} \\ \hat{\mathbf{a}}_t &= \hat{\mathbf{a}}_t^- + \mathbf{K}_t \left[\begin{pmatrix} \mathbf{y}_t - f(\hat{\mathbf{z}}_t, \hat{\mathbf{a}}_t) \\ 0 - (p - f(\mathbf{p}, \mathbf{a}_t)) \end{pmatrix} \right] \\ \mathbf{P}_{\mathbf{a}_t} &= (\mathbf{I} - \mathbf{K}_t \mathbf{C}_t) \mathbf{P}_{\mathbf{a}_t}^-\end{aligned}$$

where

$$\mathbf{C}_t = \begin{pmatrix} \left. \frac{\partial f(\mathbf{z}, \mathbf{a})}{\partial \mathbf{a}} \right|_{\hat{\mathbf{a}}_t} \\ \left. \frac{\partial f(\mathbf{p}, \mathbf{a})}{\partial \mathbf{a}} \right|_{\hat{\mathbf{a}}_t} \end{pmatrix}$$

and

$$\mathbf{R}' = \begin{pmatrix} R & 0 \\ 0 & R_p \end{pmatrix}$$

The time update equations remain unchanged.

4. Examples

In this section we will examine the above ideas with application to two systems which exhibit nonlinear dynamics and chaotic behavior for certain sets of parameter values. The two systems are the well studied Chua's circuit [Madan, 1993; Kennedy, 1993] and a more recent set of equations developed by Chen [Chen & Ueta, 1999]. The Chua circuit and Chen's system can both be represented by three-dimensional systems of ordinary differential equations and the system state at time t is given by (x_t, y_t, z_t) . We will observe the x_t component of the state subject to observational noise at levels described below and attempt to reconstruct predictive models of the circuits from these time series. A portion of each time series will be used to estimate UFP's of each system to be used in the KFC algorithm.

A major problem in nonlinear dynamical system reconstruction from time series data is the question of appropriate model size. The KFC as described is not set up to answer this problem of model selection. Indeed, in [Walker & Mees,

1998] it is recommended that the extended Kalman filter be used as a tool for refining parameter estimates after application of a model selection algorithm such as the method described in [Judd & Mees, 1995]. The main purpose of this paper is to convey the idea and benefits to reconstruction of including dynamical information in the estimation process. The purpose is not to develop another model selection algorithm. As such we will perform an initial screening of the data by reconstructing radial basis models with increasing numbers of centers determined by a k-means algorithm (see, for example [Braga *et al.*, 2002]¹). The scale parameter δ of the radial basis functions was set to 1. For each model we estimate the parameters (weights) using least squares and calculate two information criteria: Akaike's information criterion (AIC) [Akaike, 1974] and the Schwarz information criterion (SIC) [Schwarz, 1978]. We will select a model size for use in the KFC which lies close to the minimum of the information criteria if one can be found. In this way we can “brush under the carpet” the model selection problem and concentrate on the KFC algorithm itself.

4.1. Chua's circuit

The Chua circuit can be represented by the following equations [Madan, 1993; Kennedy, 1993]:

$$\dot{x} = a(y - x - \phi(x))$$

$$\dot{y} = x - y + z$$

$$\dot{z} = -by$$

where $\phi(x) = m_1x + ((m_0 - m_1)/2)(|x + 1| - |x - 1|)$. The fixed points are given by $(x^*, y^*, z^*) = (0, 0, 0)$ and $(x^*, y^*, z^*) = (r, 0, -r)$ with $r = \pm((m_0 - m_1)/(1 + m_1))$. A double scroll attractor can be generated using the parameter set ($m_0 = -8/7$, $m_1 = -5/7$, $a = 0$, $b = 14.3$). Figure 1 shows a time series of the x variable together with the values of the three fixed points. (We integrated the Chua circuit equations with initial condition $(-1, 0, 0)$ from $t = 0$ s to $t = 2000$ s sampling every 0.1s and then retaining the last 3000 data points.) We consider two noise levels; a low noise case where the additive noise to the observed signal is distributed as $N(0, 0.05)$, and a slightly higher noise case with noise distributed as $N(0, 0.1)$.

A calculation of mutual information suggests a time delay lag of 6 and false nearest neighbors

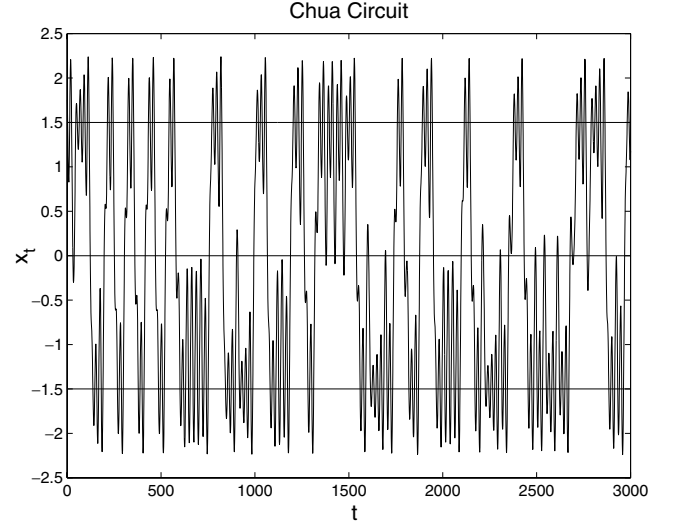


Fig. 1. x_t versus t for the Chua circuit. The horizontal lines indicate the location of the (unstable) fixed points.

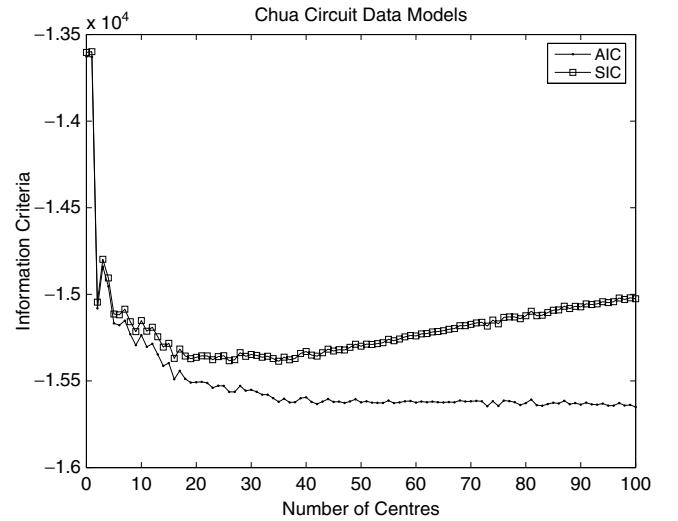


Fig. 2. AIC and SIC for initial screening of nonlinear models of the Chua circuit.

with the selected lag suggests a three-dimensional embedding [Abarbanel, 1995]. In Fig. 2 we show the results of the screening process outlined above. We see that the SIC is a minimum for a model size somewhere between 15 and 40 basis functions with the actual minimum of the calculation occurring at 35. We therefore choose to reconstruct models with 35 basis functions using the KFC.

The fixed point algorithm of Glover and Mees modified for neighbors in space suggests two definite fixed points with the possibility of a third

¹For a matlab implementation see <http://www.cc.gatech.edu/~dellaert/html/software.html>.

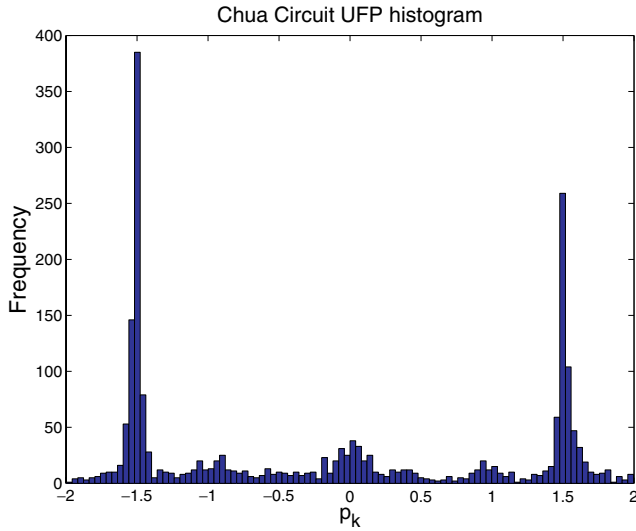


Fig. 3. Histogram of p_k for the Chua circuit data as determined by the Glover and Mees UFP locator method using neighbors in space.

at the origin (see Fig. 3). These fixed points can be seen to coincide with the true fixed points of the Chua circuit. We will apply the KFC algorithm in three cases: (i) only the UFP at the origin is used as

a constraint, (ii) the two definite UFP's at $-r$ and r suggested by the algorithm, and (iii) all three UFP's are used as constraints. The noise distribution representing the uncertainty in the UFP determination is given by the distribution $N(p, 0.1)$. We will compare the resulting model's ability to capture the local behavior corresponding to the constraints with the behavior predicted by model's reconstructed with the ordinary EKF and the screening model estimated by least squares.

In both cases the noise term associated with the models fit error is set to $N(0, 2\sigma_y^2)$ where σ_y is the standard deviation of the time series data. The initial parameter weights are set to zero except for one so that the initial model is given by $y_{t+1} = y_t$. We also only carry out one training epoch, i.e. the data is presented to the Kalman filter algorithms once.

The results for both noise cases are summarized in Tables 1 and 2. A few remarks with respect to these tables are noteworthy. Firstly since we are only optimizing over the pseudo-linear weight parameters the Kalman filter model predictions root mean square (RMS) errors are higher than

Table 1. Model errors and accuracy of fixed point estimation for the Chua circuit in the low noise case. We see again that despite the overall fit statistics — root mean square (RMS), Schwarz information criteria (SIC) and innovation errors (IE) — being worse, the KFC $(-r, 0, r)$ model estimates the dynamic properties (UFP) of the system better. See Fig. 4 for a visual representation of this improvement.

	LS	EKF	KFC (0)	KFC $(-r, r)$	KFC $(-r, 0, r)$
RMS	0.072	0.076	0.076	0.078	0.078
SIC	-1.54×10^4	-1.51×10^4	-1.51×10^4	-1.50×10^4	-1.50×10^4
IE	—	0.079	0.079	0.081	0.081
$ f(0) - 0 $	3.76×10^{-3}	5.3×10^{-3}	9.05×10^{-6}	0.006	9.6×10^{-6}
$ f(-r) - (-r) $	0.03	0.04	0.04	1.0×10^{-4}	1.0×10^{-4}
$ f(r) - r $	0.02	0.03	0.03	1.2×10^{-4}	3.9×10^{-5}

Table 2. Model errors and accuracy of fixed point estimation for the Chua circuit in the higher noise case. We see again that despite the overall fit statistics — root mean square, Schwarz information criteria and innovation errors — being worse, the KFC $(-r, 0, r)$ model estimates the dynamic properties (UFP) of the system better. See Fig. 5 for a visual representation of this improvement.

	LS	EKF	KFC (0)	KFC $(-r, r)$	KFC $(-r, 0, r)$
RMS	0.143	0.146	0.146	0.147	0.147
SIC	-1.13×10^4	-1.12×10^4	-1.12×10^4	-1.11×10^4	-1.11×10^4
IE	—	0.150	0.149	0.150	0.150
$ f(0) - 0 $	0.01	0.006	9.53×10^{-6}	5.67×10^{-3}	9.68×10^{-6}
$ f(-r) - (-r) $	0.019	0.041	0.041	1.15×10^{-4}	1.16×10^{-4}
$ f(r) - r $	0.013	0.037	0.037	1.30×10^{-4}	1.30×10^{-4}

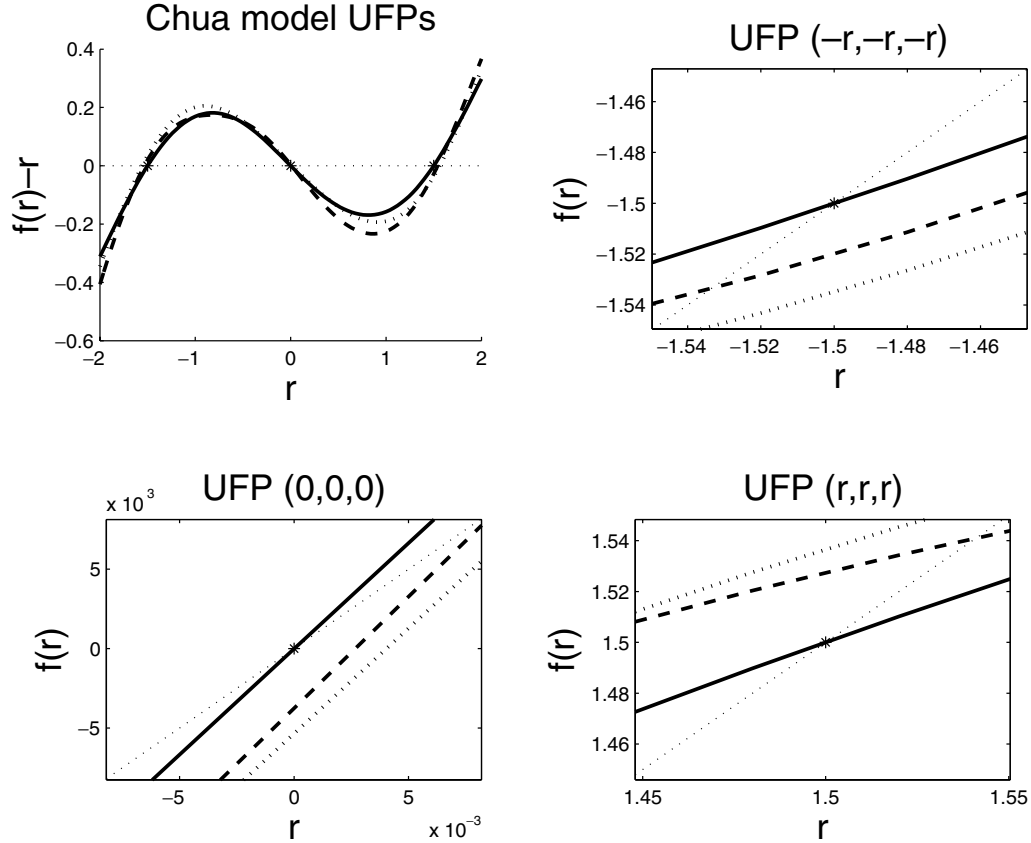


Fig. 4. Accuracy of Chua model fixed points in the low noise case. The top left panel shows a plot of the difference $f(\mathbf{r}) - r$ versus r for the models obtained using least squares (dashed), EKF algorithm (dotted) and the KFC algorithm with three fixed point constraints (solid). The remaining three panels are close up views of the fixed point accuracy (UFP's are indicated by stars). We see that the model estimated using the KFC gives much improved estimates of the UFP's over the EKF and least squares models. See Table 1 for a numerical measure of this deviation. (The lightly dashed line in these remaining panels is the diagonal $r = r$.)

the optimal least squares method (LS). It is also interesting to note that KFC RMS errors are also larger than the EKF errors. This increase in RMS is also reflected in the SIC values since the number of model parameters do not change. The accuracy of the LS and EKF models with respect to approximating the UFP's is rather poor. The performance of the KFC models with respect to this measure clearly demonstrates the benefits of incorporating the UFP's as constraints. The KFC (0) model accurately models the fixed point at the origin but not the other two. The KFC $(-r, r)$ model approximates the $\pm r$ fixed points very well but the origin is predicted poorly. Meanwhile the KFC $(-r, 0, r)$ model is able to approximate all UFP's extremely accurately with only a small loss in overall RMS error. Figures 4 and 5 give a view of the fixed point accuracy of the local accuracy of the LS, EKF and KFC $(-r, 0, r)$ models. We see that the results persist from the low to higher noise cases.

4.2. Chen's equations

Chen's system can be described by the following equations [Chen & Ueta, 1999]:

$$\begin{aligned}\dot{x} &= a(y - x) \\ \dot{y} &= (c - a)x - xz + cy \\ \dot{z} &= xy - bz\end{aligned}$$

The fixed points are seen to be given by $(x^*, y^*, z^*) = (0, 0, 0)$, $(x^*, y^*, z^*) = (r, r, r^2/b)$ and $(x^*, y^*, z^*) = (-r, -r, r^2/b)$ with $r = \sqrt{2cb - ba}$. A chaotic attractor can be generated using the parameter set $(a, b, c) = (35, 3, 28)$. Figure 6 shows a time series of the x variable together with the values of the three fixed points. (We integrated Chen's equations with initial condition $(-1, 0, 0)$ from $t = 0$ s to $t = 200$ s sampling every 0.01s and then retaining the last 3000 data points.) As in the Chua circuit example we consider two noise levels; a low noise case where the additive noise to the observed

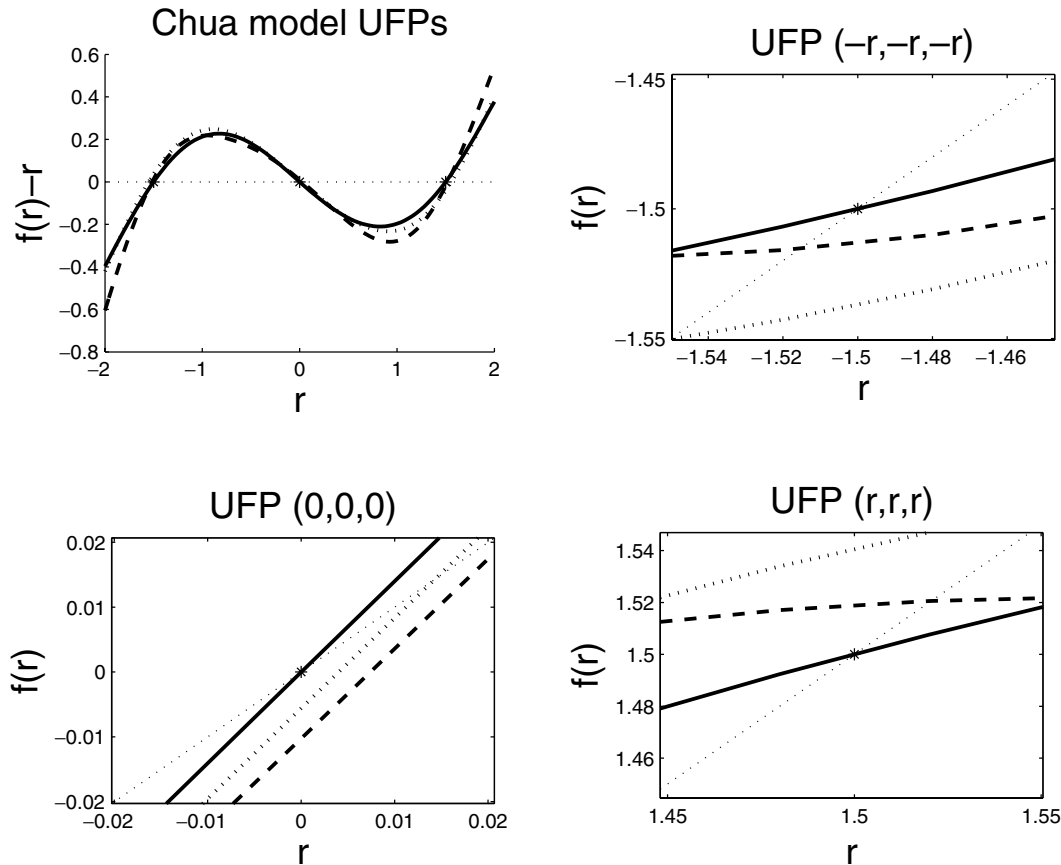


Fig. 5. Accuracy of Chua model fixed points in the higher noise case. The top left panel shows a plot of the difference $f(\mathbf{r}) - r$ versus r for the models obtained using least squares (dashed), EKF algorithm (dotted) and the KFC algorithm with three fixed point constraints (solid). The remaining three panels are close up views of the fixed point accuracy (UFP's are indicated by stars). We see that the model estimated using the KFC gives much improved estimates of the UFP's over the EKF and least squares models. See Table 2 for a numerical measure of this deviation. (The lightly dashed line in these remaining panels is the diagonal $r = r$.)

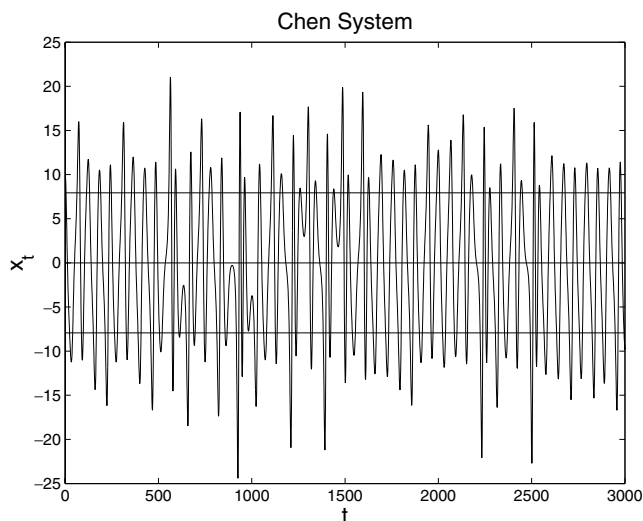


Fig. 6. x_t versus t for Chen's equations. The horizontal lines indicate the location of the (unstable) fixed points.

signal is distributed as $N(0, 0.5)$, and a slightly higher noise case with noise distributed as $N(0, 1)$.

A calculation of mutual information suggests a time delay lag of 10 and false nearest neighbors with the selected lag also suggests a three-dimensional embedding [Abarbanel, 1995]. In Fig. 7 we show the results of the screening process. The SIC is a minimum for model sizes somewhere between 60 and 140 basis functions with the actual minimum of the calculation occurring at 103. We therefore choose to reconstruct models with 103 basis functions using the EKF and KFC.

The fixed point algorithm of Glover and Mees modified for neighbors in space suggests two possible fixed points (see Fig. 8). These fixed points can be seen to coincide with the true fixed points of Chen's equations but the accuracy is a lot less certain than the definitiveness found in the Chua

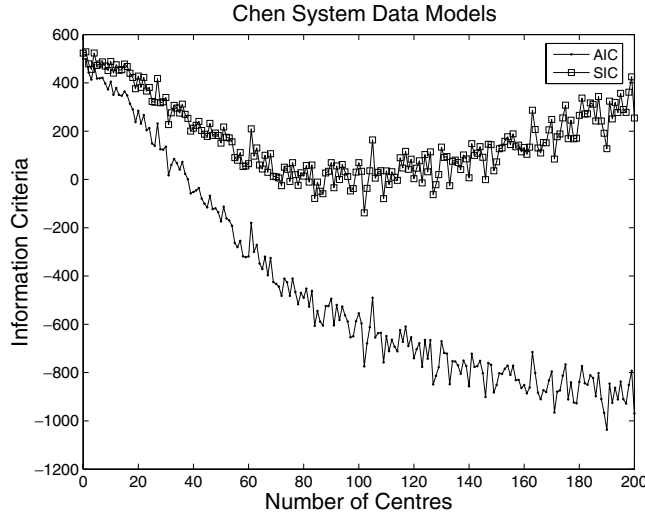


Fig. 7. AIC and SIC for initial screening of nonlinear models of Chen's equations.

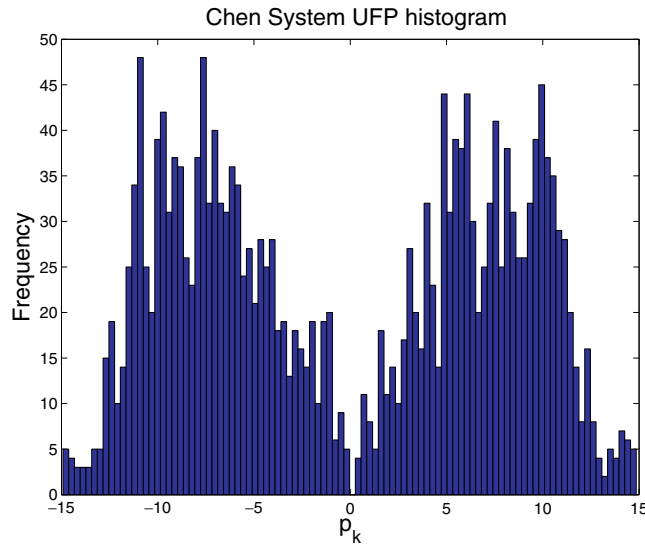


Fig. 8. Histogram of p_k for data from Chen's equations.

example. Perhaps because of this uncertainty the algorithm (subject to the embedding strategy followed) was unable to detect the UFP at the origin.

We will again apply the KFC algorithm in three cases: (i) only the UFP at the origin is used as a constraint, (ii) the two definite UFP's at the locations $-r$ and r suggested by the algorithm, and (iii) all three UFP's are used as constraints. The noise distribution representing the uncertainty in the UFP determination is given by the distribution $N(0, \sqrt{2})$. In both cases the noise term associated with the model's fit error is set to $N(0, 2\sigma_y^2)$ where σ_y is the standard deviation of the time series data. The initial parameter weights are set to zero except for one so that the initial model is given by $y_{t+1} = y_t$. We also only carry out one training epoch, i.e. the data is presented to the Kalman filter algorithms once.

The results for both noise cases are summarized in Tables 3 and 4. The results are very similar to those found for the Chua circuit and so the remarks with respect to RMS and SIC apply here. We see that the accuracy of the LS and EKF models with respect to approximating the UFP's is rather poor. The performance of the KFC models with respect to this measure clearly demonstrates the benefits of incorporating the UFP's as constraints. The KFC (0) model accurately models the fixed point at the origin but not the other two. The KFC $(-r, r)$ model approximates the $\pm r$ fixed points very well but the origin is predicted poorly. Meanwhile the KFC $(-r, 0, r)$ model is able to approximate all UFP's extremely accurately with only a small loss in overall RMS error. Figures 9 and 10 give a view of the fixed point accuracy of the local accuracy of the LS, EKF and

Table 3. Model errors and accuracy of fixed point estimation for Chen's equations in the low noise case. We see that despite the overall fit statistics — root mean square, Schwarz information criteria and innovation errors — being worse, the KFC $(-r, 0, r)$ model estimates the dynamic properties (UFP) of the system better. See Fig. 9 for a visual representation of this improvement.

	LS	EKF	KFC (0)	KFC $(-r, r)$	KFC $(-r, 0, r)$
RMS	0.861	0.894	0.895	0.934	0.935
SIC	-37.06	190.21	196.78	451.14	454.83
IE	—	0.952	0.952	0.989	0.989
$ f(0) - 0 $	0.26	0.17	7.95×10^{-5}	0.14	4.9×10^{-6}
$ f(-r) - (-r) $	1.31	0.80	0.78	3.9×10^{-4}	2.6×10^{-5}
$ f(r) - r $	1.28	0.80	0.81	5.3×10^{-4}	3.9×10^{-5}

Table 4. Model errors and accuracy of fixed point estimation for Chen's equations in the higher noise case. We see again that despite the overall fit statistics — root mean square, Schwarz information criteria and innovation errors — being worse, the KFC $(-r, 0, r)$ model estimates the dynamic properties (UFP) of the system better. See Fig. 10 for a visual representation of this improvement.

	LS	EKF	KFC (0)	KFC $(-r, r)$	KFC $(-r, 0, r)$
RMS	1.489	1.493	1.494	1.549	1.549
SIC	3234.3	3244.5	3246.4	3462.8	3463.9
IE	—	1.557	1.556	1.603	1.603
$ f(0) - 0 $	0.165	0.161	6.73×10^{-5}	0.135	5.84×10^{-5}
$ f(-r) - (-r) $	0.6	0.994	0.981	6.28×10^{-4}	6.1×10^{-4}
$ f(r) - r $	0.527	1.039	1.046	8.78×10^{-4}	8.94×10^{-4}

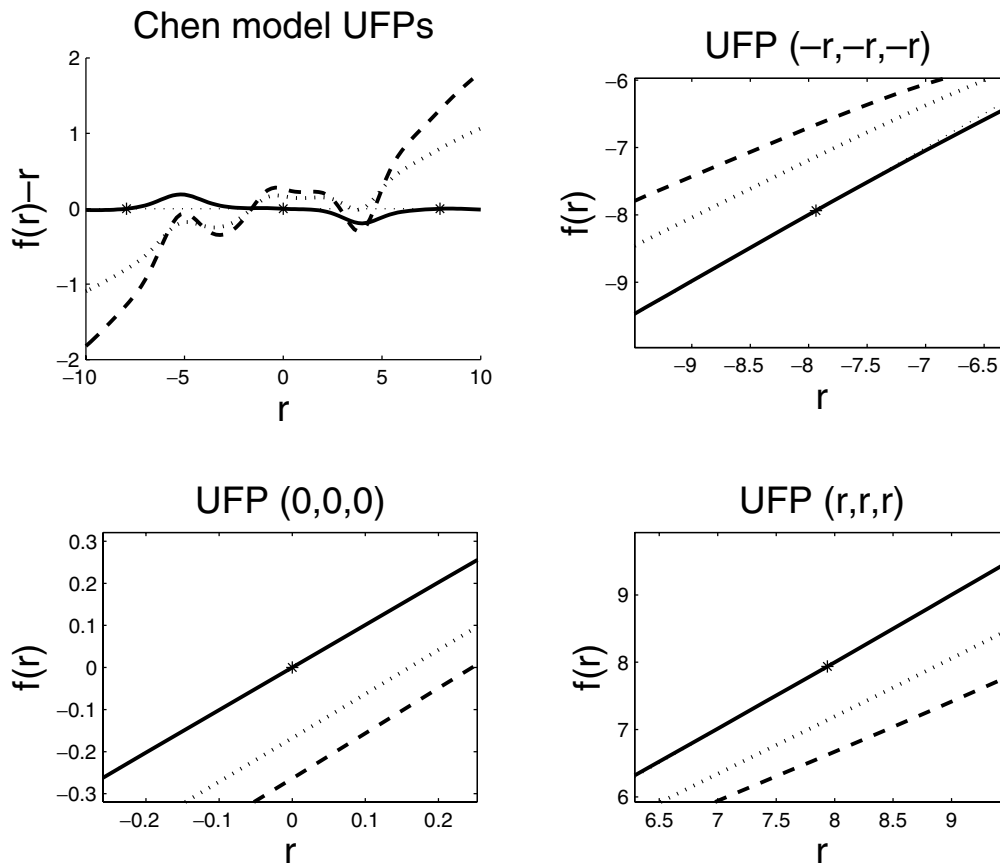


Fig. 9. Accuracy of model fixed points in the low noise case. The top left panel shows a plot of the difference $f(\mathbf{r}) - r$ versus r for the models obtained using least squares (dashed), EKF algorithm (dotted) and the KFC algorithm with three fixed point constraints (solid). The remaining three panels are close up views of the fixed point accuracy (UFP's are indicated by stars). We see that the model estimated using the KFC gives much improved estimates of the UFP's over the EKF and least squares models. See Table 3 for a numerical measure of this deviation. (The lightly dashed line in these remaining panels is the diagonal $r = r$.)

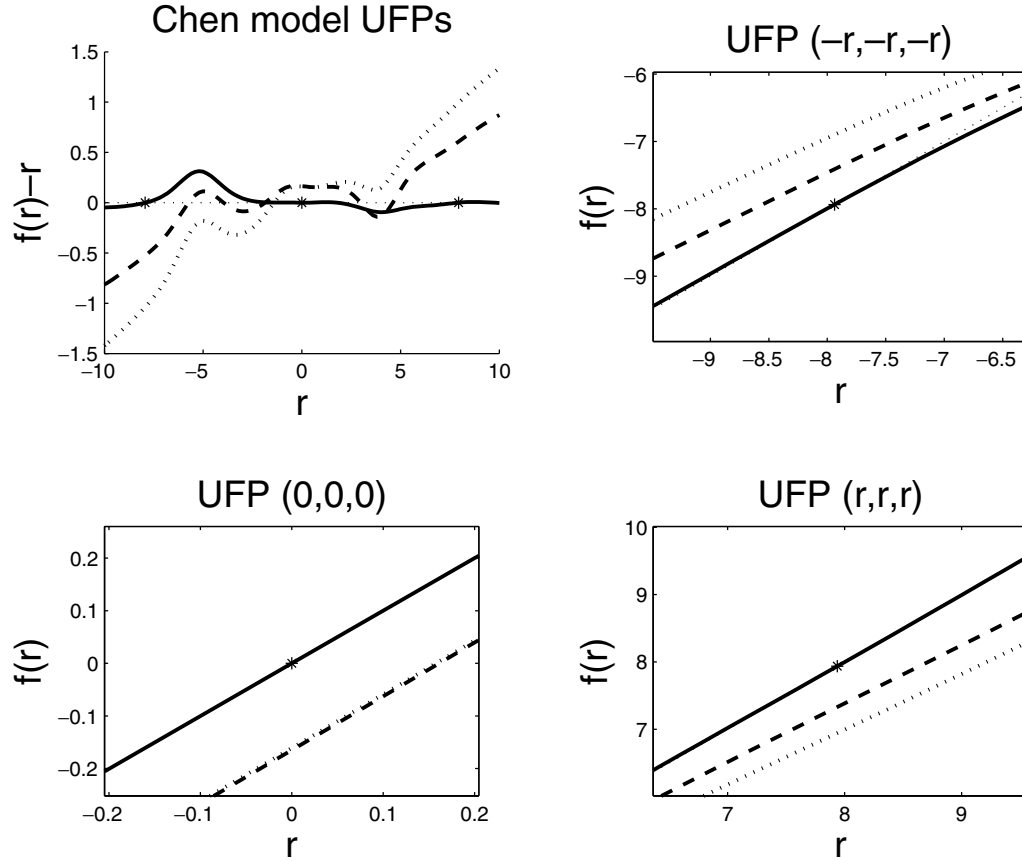


Fig. 10. Accuracy of model fixed points in the higher noise case. The top left panel shows a plot of the difference $f(\mathbf{r}) - r$ versus r for the models obtained using least squares (dashed), EKF algorithm (dotted) and the KFC algorithm with three fixed point constraints (solid). The remaining three panels are close up views of the fixed point accuracy (UFP's are indicated by stars). We see that the model estimated using the KFC gives much improved estimates of the UFP's over the EKF and least squares models. See Table 4 for a numerical measure of this deviation. (The lightly dashed line in these remaining panels is the diagonal $r = r$.)

KFC $(-r, 0, r)$ models. We see that these results also persist from the low to higher noise cases.

5. Summary

We have introduced the idea of using the fixed point structure of a dynamical system to aid parameter estimation of nonlinear models. This information can take the form of constraints in the optimization problem associated with reconstruction. The extended Kalman filter algorithm provides a natural framework in which to incorporate such constraints; one simply includes a measure of model fixed point accuracy as an extra observation function in the EKF system model. We coined this modification the KFC — Kalman Filter with Constraints — algorithm.

The unstable fixed points can be estimated from times series data and the errors in estimation can guide the noise covariances to be used in the filters. We showed using two examples exhibiting chaotic dynamics that the KFC models do indeed capture the local dynamics structure better than models reconstructed using least squares and ordinary extended Kalman filters.

It was also shown that the improved accuracy of the local behavior was only marginally detrimental to the overall global fit to the data. This could have interesting consequences for control systems where good local dynamic approximations for stabilization and more global knowledge for targeting is required.

There are some limitations to the models we have reconstructed. We have not studied the long term free-running simulation of our models. The

reason for this is that we have grossly simplified the problem by screening models to determine model size and furthermore we have only used the KFC to modify the weight parameters of these models. The EKF and KFC framework allows for all parameters to be refined which does lead to improved RMS errors and lower information criteria values [Walker & Mees, 1998].

We feel it is more important to convey the idea of including dynamical information in the estimation process as beneficial and quite straightforward if a Kalman filter is used. We suspect that the incorporation of such constraints in more sophisticated model selection and estimation algorithms would be worthwhile.

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